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relationships.

an essential cognitive skill in lows students to interpret and epresentations. Prior studies have demonstrated that visualizations enhance learning, support problem-solving, and help integrate prior knowledge with new information [3]. Tasks involving chemical representations—especially at the submicroscopic and symbolic levels—require cognitive skills such as mental rotation, spatial transformation, and visual pattern recognition [4].

Psychometric research identifies various types of spatial ability, with chemistry education emphasizing three key dimensions: spatial visualization, closure flexibility, and spatial relations [4,5]. Spatial visualization involves mentally manipulating complex objects, typically assessed using the Purdue Visualization of Rotations Test (PVRT). Closure flexibility is the capacity to quickly identify visual patterns within cluttered fields, while spatial relations concern the rapid mental manipulation of simpler shapes.

Early studies found a significant correlation between spatial ability and success in general chemistry, even in areas not traditionally considered spatial, such as solving stoichiometry problems ($r = 0.32$) [6]. This raises the question of why visual-spatial ability plays a role in non-spatial topics. Some scholars, such as Cheng and Gilbert [7], argue that understanding concepts like stoichiometry involves representational conventions rather than spatial thinking. However, to solve stoichiometry problems effectively, students must interpret chemical symbols, visualize molecular-level interactions, and apply algorithmic reasoning [34].

Stoichiometry focuses on the quantitative relationships between reactants and products in chemical reactions. For example, when interpreting the balanced equation $C(s) + O_2(g) \rightarrow CO_2(g)$, students often perceive the equation as a collection of letters and numbers instead of a representation of chemical bonds breaking and forming. Chemists, in contrast, visualize this as a dynamic process involving molecular interactions. Many students struggle to connect these representations to the underlying concepts [8]. The authors hypothesize that such difficulties may stem from differences in students' visual-spatial abilities.

Chemistry learning is abstract and difficult to understand if there is no visualization of the model by developing problem-solving skills will achieve the main goal of chemistry learning [40]. Bodner [9] emphasized that successful problem solvers can use multiple representations, especially diagrams, to conceptualize and resolve problems. Representational competence—the ability to interpret and translate between different chemical representations—is critical for success in this domain [10,11,12]. When students possess strong

understanding. Wulandari and Rusmini [31] reported that ECIRR learning models significantly reduced students' misconceptions in stoichiometry. Elvina and Latisma [32] emphasized that students' ability to comprehend chemical phenomena depends on their skill in interpreting multiple representations. Additionally, Cahyani et al. [33] demonstrated that visually rich media such as e-magazines can enhance students' interest in chemistry learning, which may support better engagement with symbolic and submicroscopic concepts.

However, using correct representations does not automatically imply conceptual understanding. Expert chemists recognize patterns and draw inferences from diagrams, while novices often rely on superficial features [16,17]. Therefore, students must learn to use chemical representations thoughtfully and accurately in problem-solving contexts [18].

To solve stoichiometry problems, students must demonstrate both conceptual and procedural understanding [19]. Conceptual understanding involves interpreting phenomena through macroscopic, submicroscopic, and symbolic lenses [20], while procedural understanding relates to applying mathematical and algorithmic steps. Misconceptions often arise when students fail to connect symbolic representations with the chemical realities they describe [35].

Unfortunately, visual-spatial skills are often overlooked in science education, despite their importance in interpreting diagrams, models, and symbolic equations [21]. Mental rotation, for example, plays a role not only in visual tasks but also in constructing meaningful internal models of chemical processes. Previous studies have largely focused on college students [6,23], with fewer investigations targeting high school learners. Grabow [24] explored the link between visual-spatial skills and stoichiometry performance but did not examine the cognitive strategies used by students.

Thus, this study investigates the correlation between high school students' visual-spatial ability, as measured by PVRT, and their ability to solve stoichiometry problems. It also aims to identify the strategies students use in solving both spatial and chemical tasks. This approach addresses gaps in the literature and highlights the potential of visual-spatial skill development to improve chemistry learning outcomes [36].

2. Results and Discussion

The statistical results of the visual-spatial ability variable are shown in Table 1. The subjects are 70 students with a score of PVRT minimum of 2 and maximum of 18 ($M = 9.929$; $SD = 4.041$).

Table 1. Descriptive statistics of PVRT results, including minimum, maximum, mean, and standard deviation of visual-spatial scores among the 70 participants.

Variable	N	Range	Mean	Std. Deviation	Minimum	Maximum
PVRT	70	16	9.929	4.041	2	18

The authors categorized the data as follows: students who scored ≤ 6 were considered to have low visual-spatial ability.

Those who scored between 7-13 were considered to have intermediate visual-spatial ability. Furthermore, those who

g PVRT problems includes both tal rotation. Individuals with high isual-spatial ability report that they then mentally simulate rotation in their thoughts during the first and second stages and decide on the third stage. Based on interviews, it has been determined that all students solve PVRT problems in the order of the first picture, the second picture, and the third picture (see Figure 4, from top to bottom), and no students have been found to solve the problems in a back-and-forth manner.

Individuals with higher visual-spatial abilities tend to focus more on visualizing, observing, and simulating. In comparison, those with lower visual-spatial ability tend to rely more on mentally manipulating objects to solve problems with aids. This is reflected in interview responses, where individuals with lower visual-spatial ability reported using their hands to simulate rotations. Those with intermediate visual-spatial abilities did in between. In the third picture, students with high and low visual-spatial ability make choices and comparison options. However, those with higher visual-spatial ability were observed to compare the position of the object at the beginning and the end, while the students with intermediate

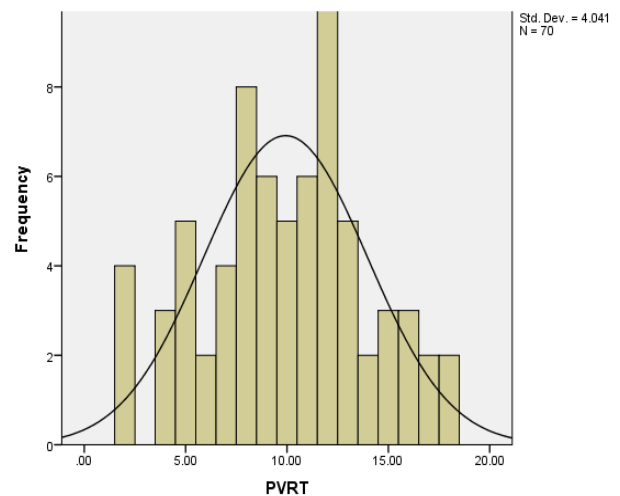


Fig. 1. Distribution of PVRT scores among 70 high school students. The bar chart shows the frequency of students falling within low, intermediate, and high visual-spatial ability categories.

Table 2. Sample student responses from the PVRT interview. This table categorizes the mental strategies used to solve mental rotation problems according to the students' visual-spatial ability levels.

Question: "How did you solve this problem?"		
Spatial Visual Ability Level	Answers	Results
High	I imagined how the part of the box would be if I rotated it. I <u>observed the shape of the plane</u> , then I <u>rotated it</u> .	Visualization, observing, simulating rotation in mind
	I <u>rotated it</u> like the first so that this small plane was not visible. I didn't observe the angle. I <u>observed it from the plane</u> just to be sure.	Simulating rotation in mind, observing
	<u>This is an example of this</u> (showing the first picture). <u>When it is rotated backward</u> , it becomes like that (showing the second picture), <u>following the hands going around</u> . If what was originally reversed like this, the <u>end is also reversed like this</u> . It depends on the opposite (plane).	Observing, simulating rotation in mind, rotating with aids, comparing item options
Intermediate	I observe the plane and how it rotates. I <u>observe the plane</u> that appears, then I <u>rotate it</u> (rotating with hand).	Observing, simulating rotation in mind
	I <u>observe from the plane then the rotation</u> . I <u>observed the top</u> . So, if the top is here, the bottom must also move here. I <u>observed the unique plane</u> .	Observing, simulating rotation in mind, rotating with aids
		Observing, simulating rotation in mind
Low	Initially <u>it's rotated once</u> following the initial. The <u>lower ones are also rotated</u> (using hand).	Observing
	Logically, if it's like this, it means that <u>another quarter circle is then rotated</u> . Another way is that I <u>twist my body</u> .	Simulating rotation in mind, rotating with aids
	If it's like this, turn it to the right side. In the example, <u>it is rotated to the right</u> . <u>This means this one is also rotated to the right</u> (using hand).	Simulating rotation in mind, rotating with aids

displays the means and standard deviation for the ability in solving stoichiometry problems. The

subjects are 70 students with a score of stoichiometry test minimum of 0.5 and maximum of 20 (M = 9.193; SD = 5.146).

Table 4. Descriptive statistics of stoichiometry test scores. The table summarizes the performance of students in solving stoichiometry problems, including score ranges and central tendencies.

Variable	N	Range	Mean	Std. Deviation	Minimum	Maximum
Stoichiometry	70	19.5	9.193	5.146	0.5	20

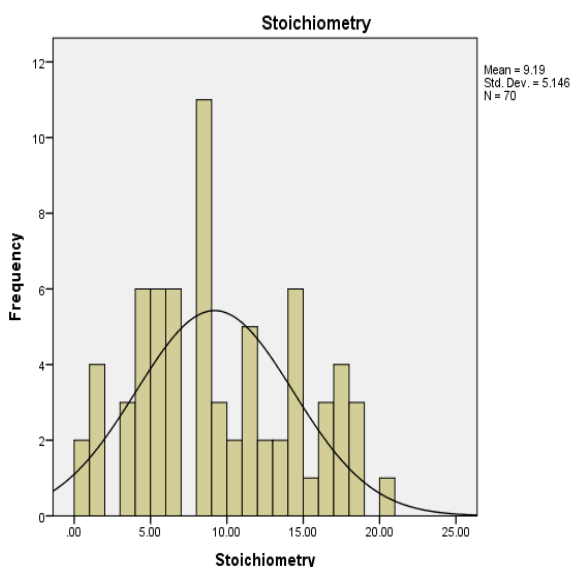


Fig. 2. Distribution of stoichiometry test scores. This histogram displays the performance levels of students on stoichiometric problem-solving tasks.

shows that the data is distributed normally.

Table 6 demonstrates the proficiency of students in solving stoichiometry problems. Question 2 on the test was the easiest for students, while question 7 was the most challenging. Questions 5 and 7, which incorporate numerical concepts, presented difficulty for students. Conversely, questions 2, 6B, and 6A, which focus on symbolic concepts, were the least challenging.

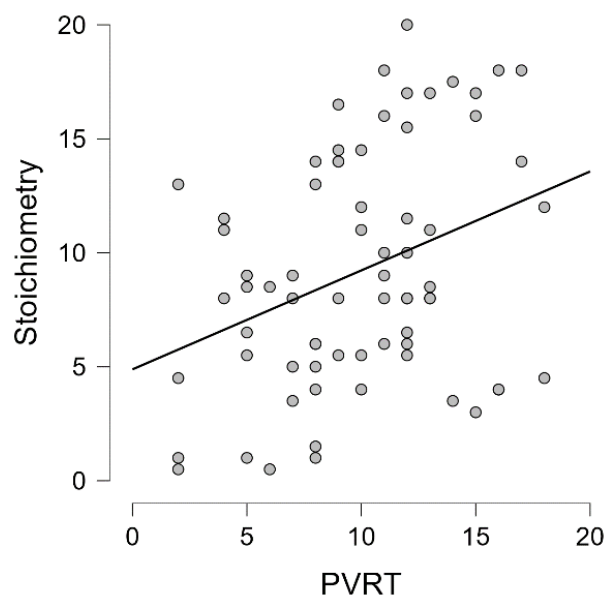


Fig. 3. Scatter plot showing the correlation between PVRT and stoichiometry test results. A trend line illustrates the positive linear relationship ($r = 0.341$, $p < 0.01$).

Table 5. Correlation analysis between PVRT and stoichiometry test results. It shows Pearson's r , p -value, and the Shapiro-Wilk test to assess normality and significance.

	Pearson's Correlations			Shapiro-Wilk Test for Bivariate Normality		
	n	Pearson's r	p	PVRT - Stoichiometry	Shapiro-Wilk	p
PVRT - Stoichiometry	70	0.341**	0.002	0.980		0.346

Note. All tests one-tailed, for positive correlation. * $p < .05$, ** $p < .01$, *** $p < .001$, one-tailed.

The study found that the easiest-hardest for the sample is symbolic-submicroscopic-symbolic-numeric. The questions involving symbolic representation (question 2) came earlier than expected, followed by questions involving submicroscopic representation (questions 6B, 6A, 1A, 4A,

solving skills (numbers 5 and 7) appeared last, as predicted. We found question 3, which should be the foundation of higher concepts, comes much later. We outline problem-solving techniques in stoichiometry problems varying from students' answers.

answer the questions in this order (according to the complexity): 1A-1B-3-6A-6B-2-4A-4B-5-7.

dent interviews on stoichiometry test items. Responses are analyzed based on the degree of correctness and representations.

Question: "How did you solve this problem?"			
No	Type of Answer	Answer	Result
Sub-microscopic representation of molecules-uni molecules			
1A	Correct	First, I look at the picture, count the bonded atoms, and count the number of molecules present. There are six molecules composed of three H atoms and one N atom.	Understand the sub-microscopic representation, change the information obtained into symbolic representation, and apply the writing rules
	Partially correct	In the picture, there are N atoms and H atoms. One N atom bonds with three H atoms, so the answer is NH_3 .	Understand the sub-microscopic representation, change the information obtained into symbolic representation, and apply the writing rules, but miss the number of molecules shown
	Incorrect with misconception	In the picture, there are N atoms and H atoms. N molecules bond with three H atoms to form NH_3 . There are six molecules, so the answer is $(\text{NH}_3)_6$.	Understand the sub-microscopic representation, and change the information obtained into symbolic representation, but do not apply the writing rules
	Incorrect	I write down the number of atoms. The picture has six black circles and 18 white circles, so the answer is N_6H_{18} .	See the sub-microscopic representation but do not understand the concept
Sub-microscopic representation of molecules-different molecules			
1B	Correct	First, I look at the picture, count the bonded atoms, and count the number of molecules present. There are five molecules consisting of one molecule of N_2 ; the N atom is covalently bonded to another N atom. In one H_2 molecule, the H atoms are covalently bonded to another H atom. And three molecules of NH_3 , one N atom bonds with three H atoms.	Understand the sub-microscopic representation, change the information obtained into symbolic representation, and apply the writing rules
	Partially correct	There are three N (in which each of them) that bonds to H. There are also three H that bonds to one N. I ignore the others.	Understand the sub-microscopic representation, change the information obtained into symbolic representation, and apply the writing rules, but miss number of molecules is shown
	Incorrect with misconception.	In the picture, there is an N molecule and an H molecule. N molecule bonds with 3 H molecules to form NH_3 . There are 3 molecules, so the answer is $(\text{NH}_3)_3$.	Understand the sub-microscopic representation, and change the information obtained into symbolic representation, but do not apply the writing rules
	Incorrect	I write down the number of atoms. The picture has five black circles and 11 white circles, so the answer is N_5H_{11} . I ignore the image's shape.	See the sub-microscopic representation but do not understand the concept
Symbolic representation of chemical reaction			
2	Correct	Hydrogen gas reacts with nitrogen	

	Partially correct	produce the NH_3 product. Then the product is balanced so that it becomes $\frac{1}{2}\text{N}_2(\text{g}) + \frac{3}{2}\text{H}_2(\text{g}) \rightarrow \text{NH}_3(\text{g})$.	manipulate the information, and partially apply the writing rules
	Incorrect with misconception	So first, I write down $\text{N}_2 + \text{H}_2$ to produce the NH_3 product. Then balance it so that it becomes $\text{N}_2(\text{g}) + (\text{H}_2)_3(\text{g}) \rightarrow (\text{NH}_3)_2(\text{g})$.	Understand the symbolic representation, can manipulate the information, but not apply the writing rules
	Incorrect,	I think about how NH_3 is formed. I don't see the constituents. So, I write $\text{N}(\text{g}) + 3\text{H}(\text{g}) \rightarrow \text{NH}_3(\text{g})$.	Do not understand the symbolic representation and cannot manipulate the information
Symbolic representation of molecules			
3	Correct	The coefficient indicates the number of molecules. Subscript shows the number of bonded atoms. So that 3N_2 , 2 suggests that there are two bonded N atoms, and 3 indicates that there are three N_2 molecules.	Understand the symbolic representation and can change the information obtained into sub-microscopic representation.
	Partially correct	The coefficient indicates the number of atoms in the molecule. 3 shows the number of N_2 molecules. Then the subscript does not know because it has never been taught.	Partially understand the symbolic representation and can change the information obtained into sub-microscopic representation
	Incorrect with misconception	The coefficients indicate the number of molecules, while the subscripts indicate the charge and are fixed. Usually, if the subscript is 2, it's gas. So that in 3N_2 , there are 3 molecules of N_2 , and it is a gas.	Misunderstand the meaning of coefficients
	Incorrect	The volume is equal to the molecular coefficient. While the subscript is similar to the number of bonds.	Do not understand the symbolic representation of molecules
Sub-microscopic and symbolic representation of chemical reaction			
4A	Correct	Because the reaction produces SO_3 , each sulphur atom is paired with three O atoms to form SO_3 . Since there are six S atoms and nine O_2 molecules when paired, it turns out that there are O atoms, and there is no residue to produce 6SO_3 .	Understand the representation at symbolic levels, connect, and use information obtained into sub-microscopic representation
	Partially correct.	There are five sulphur molecules and six O_2 molecules, which then react to produce five SO_3 molecules. I observe it from the S alone because the coefficient of S is the same as SO_3 , so the result is 5SO_3 .	Understand the representation at symbolic levels, connect, and use information obtained into sub-microscopic representation, but miss the number of molecules shown
	Incorrect with misconception	First, look at the reaction and then look at the picture. The reaction produces 2SO_3 . O_2 can't be separated, so to form SO_3 , there must be two bonds, which means joining. So, it starts like this (shows the picture) and reads 2SO_3 .	Understand the representation at symbolic levels, connect, and use information obtained into sub-microscopic representation, but misunderstand the meaning of coefficients and subscripts
	Incorrect	I didn't understand, so I wrote down a picture of the equation for the reaction so that there are two S, three O_2 , and two SO_3 .	Do not understand the sub-microscopic representations, cannot connect different representations
Sub-microscopic and symbolic representation of limiting reagent			

	Partially correct	There are five sulphur molecules and six O ₂ molecules, which then react to produce five SO ₃ molecules. I observe it from the S alone because the coefficient of S is the same as SO ₃ .	Understand the representation at symbolic levels, connect, and use information obtained into sub-microscopic representation, but miss number of molecules shown
	Incorrect with misconception	First, I look at the reaction, then look at the picture. The reaction produces 2SO ₃ . O ₂ can't be separated, so to form SO ₃ , two bonds must be joined, so it starts like this (see table), so it reads 2SO ₃ . The difference is from question 4A, some don't get a partner, so they are alone.	Understand the representation at symbolic levels, connect, and use information obtained into sub-microscopic representation, but misunderstand the meaning of coefficients and subscripts
	Incorrect	I don't understand, sir. So, I wrote down a picture of the reaction equation, sir. So, I paired the S with two O's to make it even.	Do not understand the sub-microscopic representations, cannot connect different representations
Sub-microscopic representation of chemical reaction			
6A	Correct	Three molecules of chlorine gas react with three molecules of hydrogen gas to produce six molecules of HCl gas. An equal reaction is not based on the number of molecules involved. Still, it is the simplest comparison of the number of molecules in the reaction. The picture is just an application of the balanced reaction equation, so I choose (e) Cl ₂ (g) + H ₂ (g) → 2HCl(g). There are three molecules with gas H ₂ and three Cl ₂ molecules, which have all reacted. Because it reacts, the Cl ₂ molecules that were previously paired break down. H ₂ also breaks down and then unites to become HCl. Because there are three Cl ₂ molecules and three H ₂ molecules, when they react, they run out to become six HCl.	Understand the representation at sub-microscopic levels, connect, and use information obtained into symbolic representation, and understand the writing rules
	Partially correct		Understand the representation at sub-microscopic levels, connect, and use information obtained into symbolic representation, but partially understand the writing rules
	Incorrect with misconception	I chose answer (b) 6Cl(g) + 6H(g) → 6HCl(g) because it fits the picture. The amount of substance on the left and right is the same.	Not understand the representation at sub-microscopic levels and not understand the writing rules
Sub-microscopic representation of limiting reagent			
6B	Correct	Different from question 6A, in question 6B, there are excess. The balanced reaction is not based on the number of molecules involved. Still, it is the simplest ratio of the number of molecules in the reaction. To answer, I use the direct reaction equation. One molecule of hydrogen gas reacts with half a molecule of oxygen gas to produce one molecule of water. Then I multiply by two to get rid of the fraction. The drawing is just an application of the balanced reaction equation.	Understand the representation at sub-microscopic levels, connect, and use information obtained into symbolic representation, and understand the writing rules
	Partially correct.	The difference is with question 6A; if the H atoms and Cl atoms are all bonded, in question 6B, they are not. There are still unbonded H atoms, so	Understand the representation at sub-microscopic levels, connect, and use

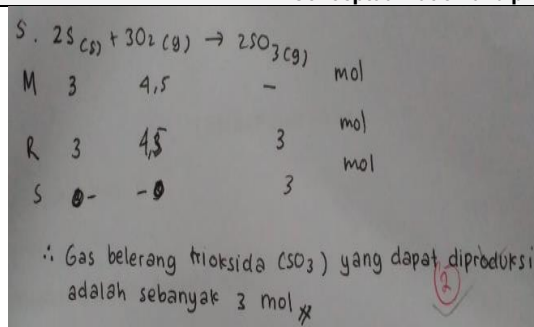
Incorrect with misconception

twelve H atoms meaning the remaining four are not used. I chose the answer (b) $(\text{O}_2)_2(\text{g}) + (\text{H}_2)_4(\text{g}) \rightarrow (\text{H}_2\text{O})_4(\text{l}) + (\text{H}_2)_2(\text{g})$ because it fits the picture. The amount of substance on the left and right is the same.

Not understand the representation at sub-microscopic levels and not understand the writing rules

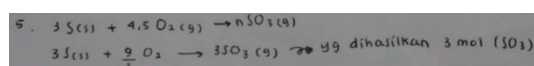
Conceptualization and problem-solving of chemical reaction

5



First, the reaction equation is balanced. Because it is equivalent, I immediately wrote. And at first, there were 3 moles of sulphur and 4.5 moles of oxygen gas then I entered them into the MRS (Initial-Change-Final) table. The reaction went perfectly. Nothing acts as a limiting reagent, so 3 moles of sulphur react with 4.5 moles of oxygen gas to produce 3 moles of SO_3 .

Understand the representation at symbolic levels, connect, use information obtained to solve numeric problem, and check whether the answer is in line with the representation

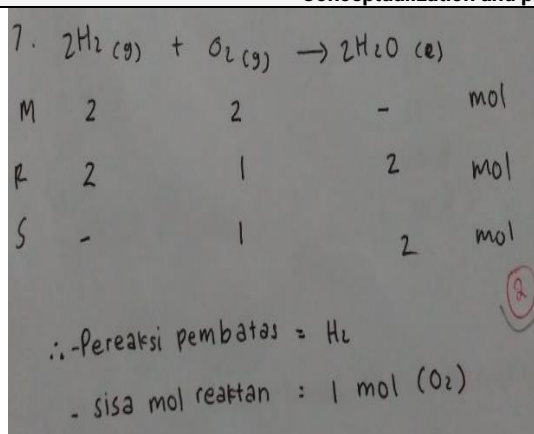


Because the product and reactant coefficients are the same, I equate the number of moles of SO_3 with the number of moles.

Understand the representation at symbolic levels, connect, use information obtained to solve numeric problem

Conceptualization and problem-solving of limiting reagent

7



First, the chemical equation is balanced to produce $\text{O}_2(\text{g}) + 2\text{H}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{l})$. And initially, there were two moles of hydrogen gas and two moles of oxygen gas. What runs out first is the limiting reagent. The one that runs out first has the largest coefficient. If hydrogen is used up first, hydrogen is the limiting reagent. So, 2 moles of hydrogen gas react with 1 mole of oxygen gas to produce 2 moles of H_2O . Since the coefficient of H_2O is 2, the remaining mole of O_2 is 1 mole.

Understand the representation at symbolic levels, connect, use information obtained to solve numeric problem, and check whether the answer is in line with the representation

Based on Table 7, individuals with correct answers tend to understand the representation (at sub-microscopic or symbolic levels), change the information into another representation, connect, use information obtained to solve (numeric) problems, and check whether the answer is in line with the representation and writing rules. Individuals with partially correct answers tend to do the same; however, they do not check or miss whether the answer is in line with the representation and writing rules. Students with incorrect answers with misconceptions tend to observe, change, connect, and use the information but misunderstand the meaning behind representation and writing rules. They tend to develop their understanding of representation differing from chemical concepts. On the other hand, students with incorrect answers only tend not to understand the representation or concept behind representation or not be able to connect different representations or not understand the writing rules.

Students are expected to answer the questions in this order (according to the questions' complexity): 1A-1B-3-6A-6B-2-4A-4B-5-7. However, students found question 2 to be the

easiest and question 7 to be the most challenging. Both questions pertained to a chemical reaction, but question 2 only dealt with a symbolic approach, while question 7 detailed numerical calculations. According to the research of Arasasingham et al. [25], students new to problem-solving in stoichiometry typically approach problems using a symbolic representation, then an algorithmic method, and finally, a visualization technique. This partially explains why question 2, which deals with a symbolic representation of a chemical reaction, is the easiest, and questions with sub-microscopic representation and numeric problem-solving come later.

The questions involving visual and molecular perspectives, specifically 6B, 6A, 1A, 4A, 1B, and 4B, were found in a sequence after those involving symbolic perspectives. This implies that students' performance on these questions may have been similar due to the need to utilize the visual molecular perspective. This order of difficulty, as found by Arasasingham et al. [25] for college chemistry students, may also apply to high school students, and it would be worth further study. Additionally, the

ultly with questions 5 and 7, which I perspective. This indicates that ith understanding and applying y may find it challenging to solve problems that require them to understand chemical phenomena, connect different representations, make changes, and transfer knowledge and skills, not just simply move among different representations [37].

What surprised us was question 3 (a symbolic representation of molecules), which comes just before 5 and 7. Understanding the coefficient and subscription in a chemical formula is the foundation of chemistry. The inability to understand and explain those can lead to misconceptions [38]. Still, students may be able to solve the stoichiometry problem by memorizing steps as mathematics equations, but this will lead to failure to understand the chemical concept behind the symbolic representation of molecules [39].

This study found evidence supporting that students who performed well on the PVRT tend to perform well on stoichiometry problem-solving tests and vice versa. This indicates that students with higher visual-spatial abilities are more likely to be successful when solving stoichiometry problems than those with lower visual-spatial abilities. This correlation is not a coincidence; visual-spatial abilities are necessary for solving stoichiometry problems. This is supported by previous research conducted by Bodner & McMillen [6] and Bodner et al. [23] in college-level settings as well as Grabow [24] in high-school-level settings.

We found a link between the abilities to solve PVRT and stoichiometry problems. Based on interviews, individuals who are better at visual-spatial tasks tend to rely more on observing, visualizing, and simulating in their minds to solve problems. In contrast, those less skilled in these tasks have problems with one (or more) of those strategies and thus tend to use their hands and other tools to manipulate objects to find solutions. In addition, those with intermediate abilities fall somewhere in between. The primary strategy to solve PVRT is observing, visualizing, and simulating the rotation in mind.

Our findings in the stoichiometry test are consistent with the idea that students who got the correct answers could comprehend the information by observing the representation, visualizing and transforming it into another representation, simulating connections between representations, and using the obtained information to solve numerical problems [25]. It is important to observe, visualize, and simulate the representation in mind of stoichiometry problems to solve them. The authors related this finding to students' mastery of representational competence. There is a significant correlation between conceptual understanding and chemical representation on stoichiometry [26] and chemical equilibrium [27].

Students who struggle with stoichiometry often have difficulty understanding the meaning behind the representation and connecting it to the underlying chemical concepts. They tend to develop their understanding of representation boundless to chemical concepts. Another factor that hinders students who perform poorly is that they misunderstand or do not even understand chemistry writing rules. Therefore, observing, understanding, connecting

contrast, those with weaker visual-spatial skills did not. Similarly, in stoichiometry tests, students who performed well were observed to carefully check if their answers adhered to the representation and writing rules. Verifying answers is essential because people with partially correct answers may perform just as well as those with correct answers. However, they may need to verify or realize that their answers are not following the representations and writing rules.

In previous research conducted examining the link between spatial thinking and success in problem-solving [30], it was determined that spatial ability appears to play a significant role in the problem conception stage, particularly for word problems. High levels of spatial ability enabled students to translate words into equations that could subsequently be solved. This can be seen from the interview for questions 5 and 7, where students who can solve the problems understand the problems by balancing the chemical reactions before going to the next mathematical operational steps. Failing to understand these steps (problem conceptions) will lead to failing the algorithmic operations.

Word problems that lack clear visuals in their description demand the use of visual-spatial abilities to visualize and solve them. This can be cognitively demanding, and individuals with solid visual-spatial abilities are better equipped to handle these challenges. To foster excellent scientists in the future, we should implement research on spatial cognition in educational practices to help students improve their verbal, mathematical, and visual-spatial abilities. This will likely result in improved performance in various tasks that require visualization and the ability to think creatively and form mental images from complex or vague problem descriptions.

Research on the correlation between spatial ability and problem-solving in first-year engineering students also showed that aside from having a substantial and significant effect on success in representation, the visual-spatial ability has no connection to mathematical abilities during the solution phase [30]. Students with limited spatial ability were more prone to mistakes in translating tasks and relational statements and choosing the correct schema for a problem. However, they found that there was no variance in mathematical ability (needed for solving the problems) between students with low and high spatial ability. This is in line with what we found where 42 students in question no 5 and 32 students in question no 7 can only conceptualize the problems without being able to solve the mathematical problems.

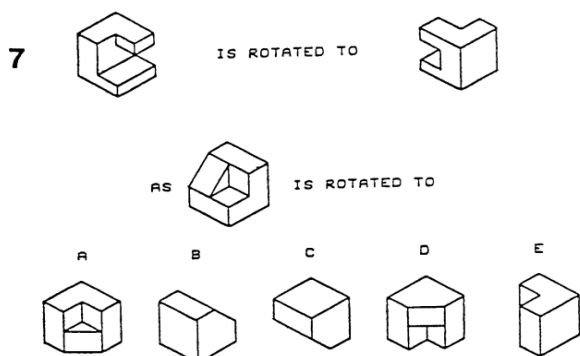


Fig. 4. Sample item from the PVRT (adapted from Wu & Shah, 2003), showing a 3D object that must be mentally rotated to determine the correct orientation among multiple-choice options.

Table 8. Classification of test questions by type of chemical representation involved (e.g., symbolic, submicroscopic). It provides context for the structure of the stoichiometry assessment.

Question code	Test information
1A	Sub-microscopic representation of molecules-uni molecules
1B	Sub-microscopic representation of molecules-different molecules
2	Symbolic representation of chemical reaction
3	Symbolic representation of molecules
4A	Sub-microscopic and symbolic representation of chemical reaction
4B	Sub-microscopic and symbolic representation of limiting reagent
5	Conceptualization and problem-solving of chemical reaction
6A	Sub-microscopic representation of chemical reaction
6B	Sub-microscopic representation of limiting reagent
7	Conceptualization and problem-solving of limiting reagent

The problems selected are stoichiometry topics, and solving them requires a thorough understanding of representations, as per the test information. It is worth

correlation. An interview was performed to clarify the students' answers and their understanding. The hypotheses of this study are

H0: There is no positive correlation between high school students' PVRT and stoichiometry test results.

H1: There is a positive correlation between high school students' PVRT and stoichiometry test results.

By conducting interviews or collecting information through open-ended questions, authors could thoroughly comprehend students' cognitive process utilized in solving PVRT and stoichiometry problems. 10 participants underwent recorded interviews where they were asked the main question: "How did you solve this problem?" The interview answers were analyzed through content analysis to extract the cognitive strategies used by the participants, which were either stated directly or deduced from the surrounding information. The authors jointly conducted the open coding analysis.

4. Conclusions

This study provides evidence of a statistically significant positive correlation between high school students' visual-spatial ability and their ability to solve stoichiometry problems ($r = 0.341$, $p < 0.01$). Students with stronger visual-spatial skills tend to better interpret representations, transform them into different formats, and apply conceptual and procedural knowledge to solve problems accurately. These findings highlight the importance of representational competence in chemistry learning and suggest that enhancing students' visual-spatial skills may support their success in topics requiring the integration of macroscopic, submicroscopic, symbolic, and numeric representations.

Limitations of this study include the relatively small and geographically limited sample, as well as the focus on only one cognitive domain. Future research should consider broader populations and explore other influencing factors such as logical-mathematical reasoning, prior knowledge, or metacognitive skills. Interventions designed to strengthen students' visual-spatial skills and their ability to navigate representational systems could be beneficial in improving conceptual understanding and problem-solving performance in chemistry.

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Author Contributions

Herunata: conceptualization, methodology, formal analysis, writing-original draft. Anugrah Ricky Wijaya:

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